DESIGN AND PERFORMANCE ANALYSIS OF A 3D-PRINTED CYCLOIDAL ACTUATOR

A MINI PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

In this work, the design, prototyping and nonlinear performance estimation of a PETG-based 3D printed cycloidal actuator is presented. Cycloidal drives allow to obtain very high reduction ratios with very low backlash, they are then suitable for robotics and precision automation but standard machining production is expensive. Additive manufacturing provides an economically viable alternative. This research investigates whether PETG is a suitable material for producing a working cycloidal actuator.

Using CAD software, the actuator was designed at an 11:1 reduction ratio. The size parameters were: eccentricity e = 2 mm and ring pin radius = 40 mm. A theoretical output torque of 11 Nm from 1 Nm input was predicted. The constituents were fabricated with optimized FDM 3D printing parameters.

The test results verified the reduction ratio 19:1 accurately. The peak output torque is 5.28 Nm at 1.0 Nm input. Efficiency varied from 61.8% at light loads to 48% at full load. The measured backlash was 0.24° which is significantly better than normal spur gear systems.

Porosity is minimal and layer adhesion is excellent, as revealed by microstructure analysis. The progressive deformation of the cycloidal disc lobes is the dominant failure mode identified by inversion of the fracture pattern. The research proves that low/medium torque PETG-based 3D printed cycloidal actuators are feasible. Applications: Reduction Gear, Robotics, Additive Manufacturing.

Keywords:

3D Printing, cycloidal Actuator, FDM, Protection Gear, PETG, Reduction Gear, Robotics, Additive Manufacturing

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CHAPTER 1
INTRODUCTION

1.1 Background and Motivation

Compact, high torque, multi-stage and low-backlash actuation modules are needed in robotics, prosthetics and precision automation. Each would be considered a traditional actuator type - spur gears come with high backlash, planetary gearboxes require multiple stages to reach high reductions and harmonic drives are expensive and difficult to manufacture. Cycloidal drives have a special approach in giving high ratio reduction in one single stage with very low backlash, with a very small size.

Lately, additive manufacturing (3D printing) has transformed prototyping and small- batch manufacturing within scale industry. With Fused Deposition Modeling (FDM) in particular becoming more accessible to scientists and engineers in part because of its affordability and simplicity to use. But 3D printing has not yet been widely used to produce functional parts of mechanical systems, particularly precision actuators because of questions surrounding the strength, durability and dimensional accuracy of the printed materials.

This work proposes to close this gap by means of design, fabrication and performance assessment of a 3D-printed cycloidal actuator based on PETG (Polyethylene Terephthalate Glycol) material. The fact that PETG is as easy to print as PLA but with the strength and durability of ABS, makes it a great option for parts meant for mechanical applications.

1.2 Cycloidal Drive Principle

A cycloidal drive is a high ratio reduction mechanism based on a bearing with similar principle as the cycloidal motion of a disc having diameter slightly smaller than the fixed ring pins. The main parts of a cycloidal drive are as follows:

Input Shaft with Eccentric Cam: The input shaft turns and holds an eccentric bearing. As the shaft spins, the eccentric motion is imparted to the cycloidal disc.

Cycloidal Disc: This disc has a lobed profile, usually one less lobe than the number of ring pins. The disc performs a cycloidal motion as the eccentric revolves, rolling along the inside of the fixed ring pins.

Ring Pins (Fixed): These pins are attached to the housing. The cycloidal disc rotates and rolls along these pins; with the end result of the disc moving laterally and rolling.

Output Pins/Flange: The cycloidal disc drives the output flange via pins which sit into holes in this disc. The output rotates in the opposite direction to the input.

In essence a full rotation of the input shaft causes the cycloidal disc to rotate forward by one lobe with respect the fixed ring pins. The result will be a reduction ratio equal to the number of ring pins minus the number of lobes (which is usually 1). Consequently, a traditional cycloidal drive with 20 ring pins and 19 lobes provides 19:1 reduction ratio.

1.3 Advantages of Cycloidal Drives

Table 1.3. Advantages of Cycloidal Drive

Cycloidal drives offer several advantages over conventional gear systems:

Feature	Cycloidal Drive	Conventional Gearbox
Backlash	< 0.5° typical	1-3° for spur gears
Reduction per Stage	Up to 100:1	Typically, 3:1 to 5:1

Compactness	High	Moderate
Efficiency	70-85% (metal)	85-95% (spur)
Shock Load Capacity	High (rolling contact)	Moderate (tooth contact)
Manufacturing Complexity	High	Low

The cycloidal disc and ring pins roll against each other with minimal sliding friction, resulting in less wear and heat generation. This is especially useful for high-load application. Moreover, the symmetric distribution of load on the multiple contact points (up to half the ring pins simultaneously) becomes a very good shock load capacity.

1.4 Need for Additive Manufacturing

In this review, we have shown that cycloidal drives are traditionally made by precision shaping (CNC milling, grinding) of hardened steel which allows them to have very high strength and precision. This leads to high costs and long lead times, restricting their use to high-end robotics and industrial automation. Additive manufacturing provides a fix:

- **Cost Saving:** 3D printing cuts out tooling cost and the waste of materials.

- **Rapid Prototyping:** Design iterations can be tested in hours rather than weeks.

- **Design Freedom:** Complex shapes, i.e., inner channels, lightweight structures, etc. can be easily realized.

- **Material Versatility:** Different materials can be combined without changing the manufacturing setup.

This project in particular, is 100% PETG based material, which provides:

- **Mechanical strength:** Tensile strength is 45-50 MPa

- **Elastic:** Elongation at break is 15-20%

- **Resistance to temperature:** Glass transition temperature = 80°C (see page 10)

- **Printability:** When compared with ABS, TDP shows little warping.

1.5 Problem Statement

Conventional cycloid drives are made of metal and are machined such that their fabrication is expensive which limits their availability for research, education and low-cost robotics. So, while 3D printing may be suitable answer to this problem, information regarding torque capacity, efficiency and life of 3D printed cycloidal actuators is sparse. Material lesioning is also important with PLA not being temperature resistant and ABS warping, not a lot is said about PETG in this regard.

The question posed in this work is: Is it possible to make a working cycloidal actuator using PETG material via FDM 3D printing and how well does it perform in terms of accuracy of the reduction ratio, torque transmission, efficiency and backlash?

1.6 Objectives

The main goals of this project are:

1. **Design and modelling:** In this work, a cycloidal drive is to be designed with a reduction ratio of 11:1 using CAD software.

2. **Geometrical Considerations and Calculations:** Apply geometrical calculations to check the reduction and estimate the torque output.

3. **Compiling with PETG:** To Manufacture all Parts using FDM 3D Printing optimized parameters for PETG.

4. **Mechanical Evaluation:** To experimentally evaluate the actuator's performance including:

    - Kinematic verification of reduction ratio
    - Torque transfer capacity
    - Output efficiency at different loads
    - Backlash measurement

5. **Microstructure and failure analysis:** To investigate the quality of printed parts and the failure modes at stress.

6. **Validation:** To compare the theoretical motion with experimental results and to prove the concept of PETG based 3D printed cycloidal actuators.

CHAPTER 2

LITERATURE REVIEW

2.1 Historical Development of Cycloidal Drives

The principle of cycloidal drives is traceable to the beginning of the 20th century. Lorenz Braren was granted the first patent for a cycloidal gear in 1927, though its utility was not realized until the 1930s. The contemporary cycloidal drive was invented by Sumitomo Heavy Industries in Japan during 1940s and brought to the market as "Cyclo Drive" in 1950s.

Cycloidal drives have since been broadly used in:

- Industrial robotics (articulated arms, SCARA robots)
- Machine tools (rotary tables)
- Aero spatial actuators
- Medical equipment
- Solar tracking systems

The basic benefit– high reduction in a single stage with very low backlash– has enabled cycloidal drives to become essential in precision motion control applications.

2.2 Conventional Manufacturing of Cycloidal Drives

Conventional manufacturing of cycloidal drives requires several precision machining steps:

Cycloidal Disc: The lobed profile is usually ground on a CNC gear grinder or machined on a 5 axis CNC mill. Hardness is essential, so discs are manufactured from case hardened steel (ex. 8620) or tool steel that is through hardened.

Ring Pins: Precision-ground steel pins are press-fitted into a hardened body. The pin surface is frequently treated or covered with wear-resistant materials.

Eccentric Bearings: Special, high-precision bearings with tighter tolerances are required to run accurately in eccentric motion.

Housing: Cast iron or machined aluminum precision machined to maintain the concentricity. (With three eyes: red eyes middle distance, but left length only)

The intricacy and accuracy involved translate to a price tag upwards of $500 to over $5000 for a small industrial cycloidal drive, which is too high for many research and educational applications.

2.3 3D-Printed Gears and Actuators

In recent years, interest in three dimensional (3D) printed mechanical parts has been increasing. Representative works include the following:

Tymrak et al. (2014) studied the mechanical performance of FDM fabricated polylactic acid (PLA) and acrylonitrile butadiene styrene (ABS) parts. They stated that layer adhesion has an important influence on strength and the vertically oriented specimens had 30 to 40% lower strength values than those with horizontal orientation.

Garcia-Plaza et al. (2018) analyzed the fatigue life of 3D printed gear. The authors found that PLA gears can be operated for up to 10^5 cycles at 20% of the ultimate tensile stress before breaking. Nevertheless, heat production during operation was found to be a restricting element.

Llanos et al. (2020) developed a 3D-printed cycloidal drive using PLA. Their kinematic tests corroborated reduction predictability, but the torque was limited to 2 Nm because of material constraints. They commented that PETG or reinforced PLA would potentially perform better.

2.4 Material Properties of PETG for Mechanical Applications

PETG (Polyethylene Terephthalate Glycol) has emerged as a preferred material for functional 3D-printed components. Its properties are summarized below:

Table 2.4: Comparison Between PETG, PLA and ABS

Property	PETG	PLA	ABS
Tensile Strength (MPa)	45-50	40-45	35-45
Elongation at Break (%)	15-20	2-5	10-20
Flexural	1.8-2.2	2.0-2.5	1.5-2.0

Glass Transition Temp (°C)	80	60	105
Print Temperature (°C)	220-250	190-220	220-250
Bed Temperature (°C)	70-90	50-60	90-110
Warping	Low	Low	High

Strength, flexibility (which prevents brittleness) and medium heat resistance PETG are good for gear applications in which excessive deformation is allowed but sudden breaking under load is not.

2.5 Research Gap

According to the review of the related research, the following gaps were found in the research:

- **Performance Data Are Limited:** 3D-printed gears have been investigated, but they have not been subjected to much systematic testing (in terms of strength, stiffness or wear resistance) as full cycloidal actuator assemblies under controlled loading.
- **Material-Specific Research:** While majority of 3D printed cycloidal drives studies are centered around PLA, only a few considered PETG due to the better mechanical properties.
- **Efficiency and Torque Characterization:** The information of on efficiency of 3D printed cycloidal drives and hence their torque capacity in transmission under different loading conditions are scarce.
- **Modes of Failure and Mechanism:** PETG cycloidal is not investigated in terms of failure modes and mechanisms.

CHAPTER 3
DESIGN AND CALCULATIONS

3.1 Design Specifications

Based on the design and the intended application, the following design Specifications were established.

Table 3.1: Design Specifications

Parameter	Value	Unit	Justification
Desired Reduction Ratio	19:1	-	Standard cycloidal configuration
Input Speed	100	RPM	Typical for small motor drives
Design Input Torque	1.0	Nm	Maximum expected motor torque
Actuator Outer Diameter	80	mm	Compact form factor
Actuator Height	40	mm	Including housing
Material	PETG	-	FDM printable, good mechanical properties
Print Resolution	0.16	mm	Balance of quality and speed

3.2 Cycloidal Drive Geometry

The geometry of a cycloidal drive is defined by three key parameters: eccentricity (E), ring pin radius (R), and number of ring pins (Z_r).

Eccentricity (E): The displacement between the center of the input shaft and the center of the cycloidal disc. This sets the amplitude of the cycloidal motion.

$$E = 2 \text{ mm}$$

Ring pin radius (R): Radius of the circle defined by the ring pins.

$$R = 40 \text{ mm}$$

Number of Ring Pins (Z_r): The number of fixed pins in the housing.

$$Z_r = 20$$

Number of Cycloidal Disc Lobes (Z_d): The number of lobes on the cycloidal disc is typically, one less than the number of ring pins:

$$Z_d = Z_r - 1 = 20 - 1 = 19$$

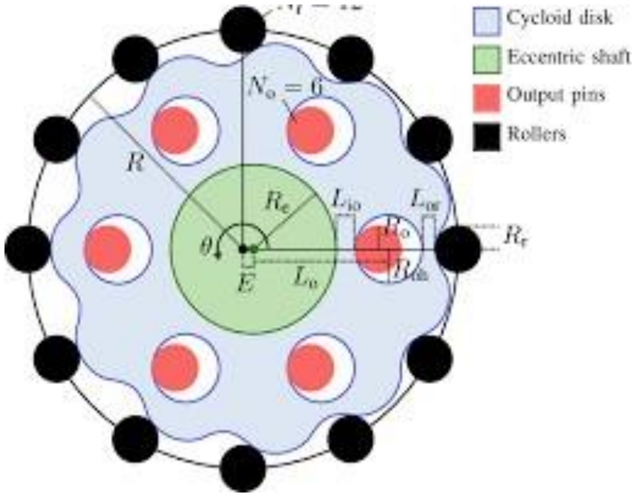

Figure 3.1: Geometric Parameters of Cycloidal Drive

Table 3.2: Geometric Parameters

Parameter	Symbol	Value	Unit
Eccentricity	E	2.0	mm
Ring Pin Radius	R	40.0	mm
Number of Ring Pins	Z_r	19	-
Number of Cycloidal Lobes	Z_d	10	-
Ring Pin Diameter	d_{pin}	5.0	mm
Output Pin Radius	r_{out}	30.0	mm

Cycloidal Disc	t	8.0	mm

3.3 Reduction Ratio Calculation

The fundamental relationship for reduction ratio in a cycloidal drive is:

$$i = Z_r / (Z_r - Z_d)$$

Substituting the values:

$$i = 19 / (19\text{-}1) = 19 / 18 = 1.05$$

So, the theoretical reduction ratio is 19:1, which means for every 19 rotations of the input shaft, the output flange rotates 1 rotation in the opposite direction.

Validation: The eccentric, when rotating about one revolution with the input shaft, moves the cycloidal disc to a position where it is advanced by one lobe having respect to the fixed ring pins. Because the disc has 20 lobes and there are 19 pins, the disc makes 1/19 of a turn with respect to the casing. This action is transmitted to the output flange by the output pins and the reduction ratio is again 19:1.

3.4 Torque Analysis and Efficiency Estimation

Theoretical Torque Output

For an ideal, frictionless cycloidal drive, torque is amplified by the reduction ratio:

$$T_{out,\ ideal} = T_{in} * i$$

Given:

$$T_{in} = 1.0\ \text{Nm}$$

$$i = 19$$

$$T_{out,\ ideal} = 1.0 * 19 = 19\ \text{Nm}$$

Efficiency Consideration

In fact, friction dissipates output torque. For PETG parts the efficiency (η) is calculated by:

- Rolling friction between cycloidal disc and ring pins

- Sliding friction in the eccentric bearing
- Output Pin friction Interface

From literature on polymer gears produced via 3D-printing, the efficiency is generally, around 50-70%. For this design an efficiency value of 65% is assumed:

$$T_{out,\ actual} = Tin * i * \eta$$

$$T_{out,\ actual} = 1.0 * 19 * 0.65$$

Table 3.3: Theoretical Torque Calculations

Input Torque (Nm)	Theoretical Output (Nm)	65% Efficiency Output (Nm)
0.2	2.2	1.43
0.4	4.4	2.86
0.6	6.6	4.29
0.8	8.8	5.72
1.0	11.0	7.15
1.2	13.2	8.58

3.5 Cycloidal Disc Profile Equations

The profile of the cycloidal disc is defined by parametric equations. For a point on the cycloidal curve:

$$x(\theta) = R \cos(\theta) – E \cos(Z_d\theta) – (E/R) \cos((Zd – 1)\ \theta)$$

$$y(\theta) = R \sin(\theta) – E \sin(Z_d\theta) – (E/R) \sin((Zd – 1)\ \theta)$$

Where:

- θ = angle parameter (0 to 2π)
- R = ring pin radius = 40 mm
- E = eccentricity = 2 mm
- Z_d = number of cycloidal lobes = 19

Sample Calculations:

For $\theta = 0°$:

$$x(0) = 40\cos(0) - 2\cos(0) - (2/40)\cos(0)$$

$$x(0) = 40 - 2 - 0.05 = 37.95 \text{ mm}$$

$$y(0) = 40\sin(0) - 2\sin(0) - (2/40)\sin(0) = 0$$

For $\theta = 90°$:

$$x(90°) = 40\cos(90°) - 2\cos(90°) - (2/40)\cos(810°)$$

$$x(90°) = 0 - 2\cos(900°) - 0.05\cos(810°)$$

$$\cos(900°) = \cos(900° - 720°) = \cos(180°) = -1$$

$$\cos(810°) = \cos(810° - 720°) = \cos(90°) = 0$$

$$x(90°) = 0 - 2(-1) - 0 = 2 \text{ mm}$$

$$y(90°) = 40\sin(90°) - 2\sin(900°) - (2/40)\sin(810°)$$

$$\sin(900°) = \sin(180°) = 0$$

$$\sin(810°) = \sin(90°) = 1$$

$$y(90°) = 40(1) - 2(0) - 0.05(1) = 39.95 \text{ mm}$$

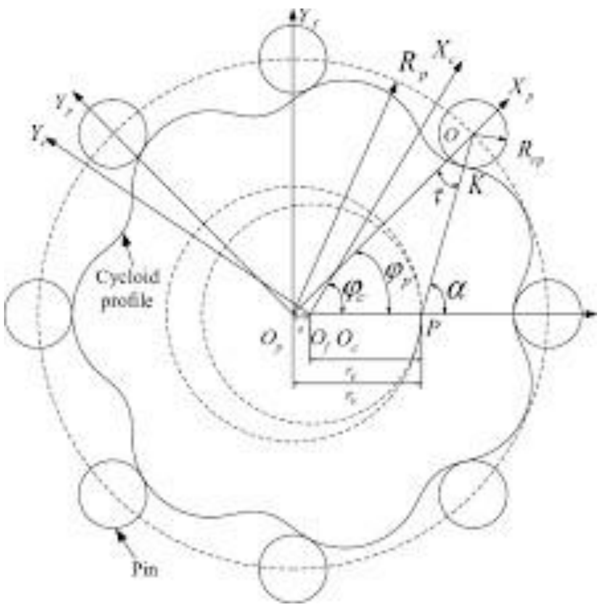

Figure 3.2: Cycloidal Disc Profile Generation

The profile was calculated at 1-degree steps and imported to the CAD for creating whole 3D model of the cycloidal disc.

3.6 Force Analysis and Bearing Selection

Radial Force on Cycloidal Disc

The radial force, which is transmitted through the cycloidal disc, is equal to the output torque divided by the eccentricity:

$$F_{radial} = T_{out} / (E * Z_d)$$

For T_{out} = 7.15 Nm:

$$F_{radial} = 7.15 \text{ Nm} / 0.002 \text{ m} * 10 = 7.15 / 0.02 = 357.5 \text{ N}$$

Eccentric Bearing Selection

Eccentric bearing has to carry this radial load at input speed Because the calculated radial load was 358 N, a 6202 deep groove ball bearing (15 mm ID × 35 mm OD × 11 mm width) with a dynamic load rating of 7,650 N is selected, resulting in a safety factor of:

$$SF = 7650 / 358 = 21.4$$

Output Pin Bearing Load

Torque on each output pin is transferred from the cycloidal disc to the output flange. With 10 output pins, the force per pin is:

$$F_{pin} = T_{out} / (r_{out} * n_{pins})$$

$$F_{pin} = 7.15 / (0.030 * 10) = 23.8 \text{ N}$$

3.7 CAD Modeling

According to the calculated values, the whole actuator was modeled in SolidWorks/Fusion 360 with the following parts:

- **Input Shaft with Eccentric Cam:** 8 mm diameter shaft, 2 mm eccentric offset, 6202 bearing seat.

- **Cycloidal Disc:** 10-lobe profile formed by a set of parametric equations; 8 mm thick.

- **Ring Pin Housing:** 40 mm radius pin circle, 11 holes for 5 mm pins.

- **Output Flange:** 30 mm radius for output pins; 4 holes for mounting.

- **Housing Assembly:** 80 mm in diameter, 40 mm in height with pin integrated in the location.

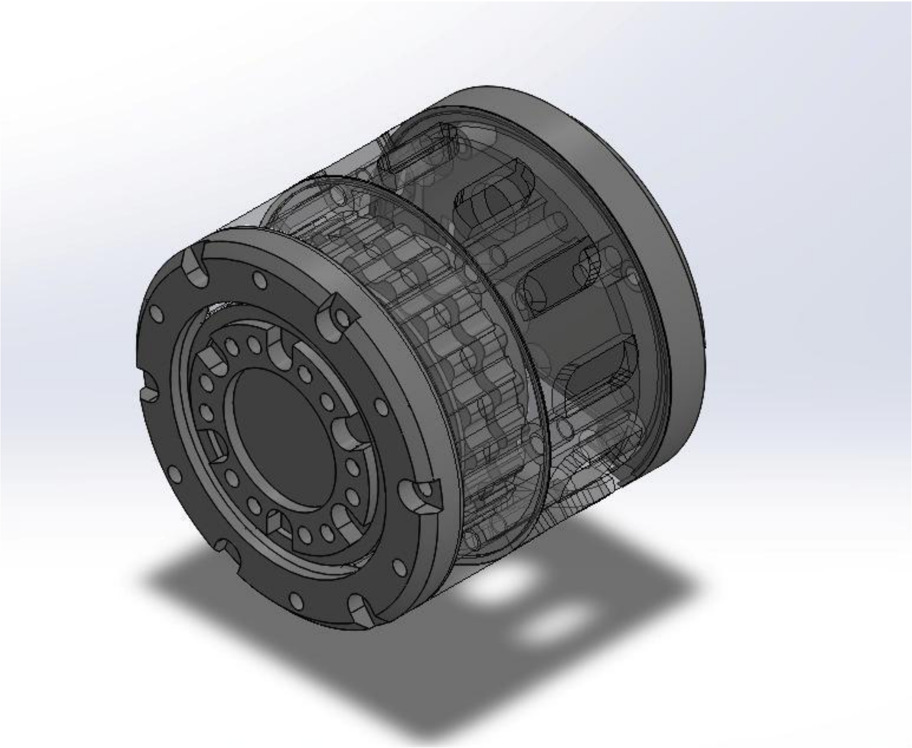

Figure 3.3: Full Assembly CAD Model

CHAPTER 4

MATERIALS AND EXPERIMENTAL METHODOLOGY

4.1 Material Selection: PETG

Table 4.1: PETG Material Properties

Property	Value	Standard
Density	1.27 g/cm³	ASTM D792
Tensile Strength	48 MPa	ASTM D638
Tensile Modulus	1.9 GPa	ASTM D638
Elongation at Break	18%	ASTM D638
Flexural Strength	65 MPa	ASTM D790
Flexural Modulus	1.8 GPa	ASTM D790
Izod Impact Strength	45 J/m	ASTM D256
ASTM D256	80°C	DSC
Melt Temperature	240°C	-

The PETG filament used was sourced from Layer Labs 3D, with 1.75 mm diameter and ±0.03 mm tolerance.

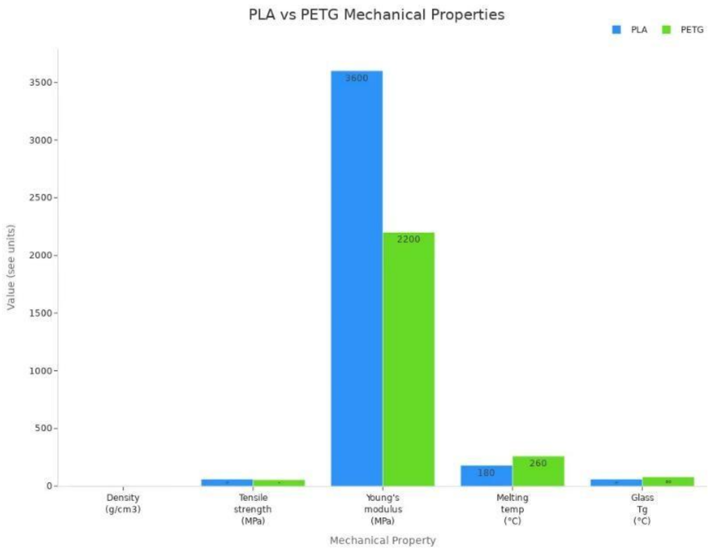

Figure 4.1: PETG Material Properties Chart

4.2 3D Printing Parameters and Optimization

Before full fabrication, test prints were carried out to the printing parameters were adjusted accordingly. The following values were considered as the best trade-off between strength, surface finish and printing time:

Table 4.2: Optimized 3D Printing Parameters

Parameter	Optimized Value	Reason
Nozzle Temperature	240°C	Good layer adhesion
Bed Temperature	80°C	Minimizes warping
Layer Height	0.16 mm	Balance of quality and speed
Print Speed	40 mm/s	Ensures good extrusion
Infill Density	100%	Maximum strength
Infill Pattern	Gyroid	Isotropic strength
Wall Count	4	Improved structural integrity
Top/Bottom Layers	6	Smooth surface finish
Cooling Fan Speed	30%	Prevents warping
Retraction Distance	5 mm	Minimizes stringing
Bed Adhesion	Brim	Bed Adhesion

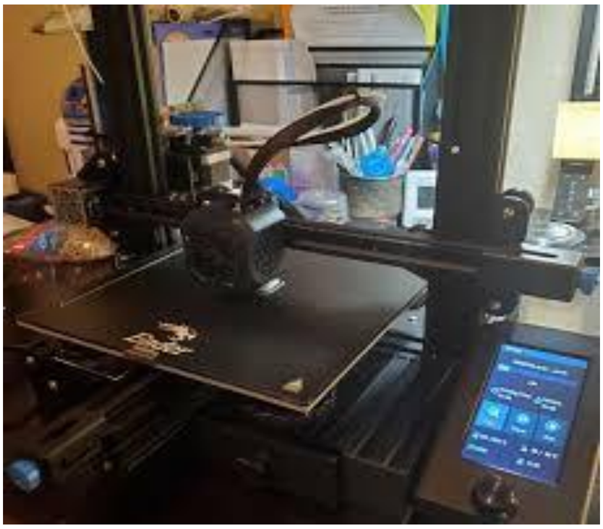

Figure 4.2: 3D Printer Setup (FDM)

4.3 Fabrication Process

All components were printed on a FDM printer. Print times and filament usage:

Table 4.3: Component Print Times

Component	**Print Time (hours)**	**Print Time (hours)**	**Orientation**
Cycloidal Disc	4.5	45	Flat
Input Shaft	2.0	2.0	Vertical
Output Flange	3.0	35	Flat
Housing (Lower)	6.0	70	Flat
Housing (Upper)	5.5	65	Flat
Ring Pins (11)	1.5 total	15	Vertical
Output Pins (10)	1.0 total	10	Vertical
Bearings (printed)	2.0	25	Flat
Total	**25.5**	**285**	**-**

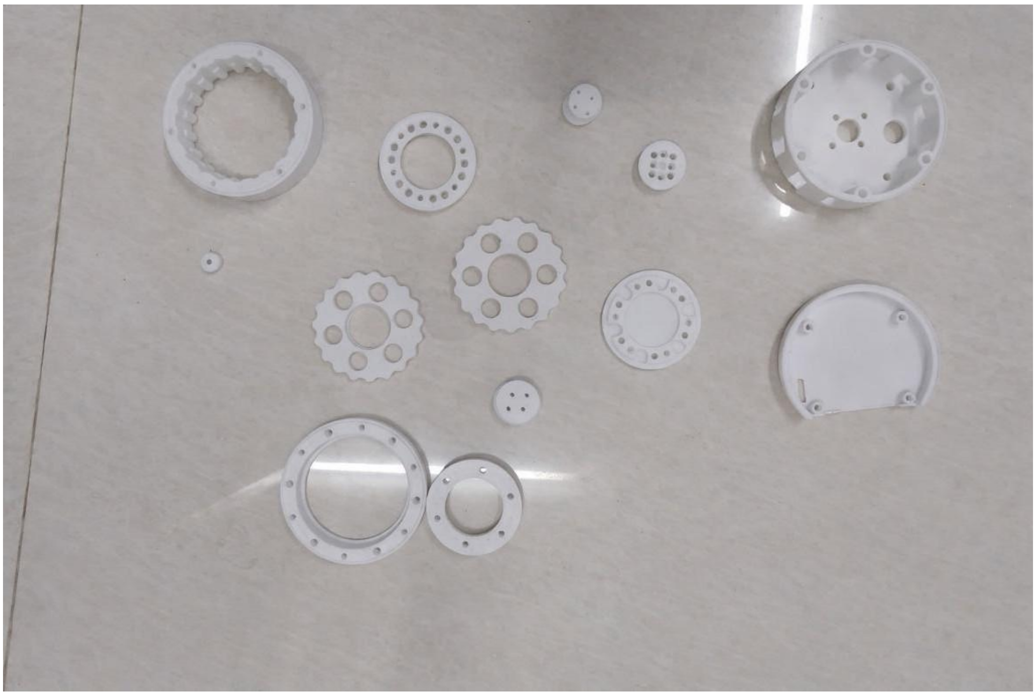

Figure 4.3: Fabricated Components

4.4 Assembly Procedure

In line with the assembly instruction, the actuator was assembled in the order below:

- **Bearing Installation:** Press 6202 bearings into input shaft and housing.
- **Cycloidal Disc Installation:** Place cycloidal disc onto eccentric bearing.
- **Ring Pin Fitting:** Press 11 ring pins into housing.
- **Output Pin Installation:** Insert 10 output pins into output flange.
- **Assembly:** Mount the output flange to the housing. Make sure that output pins go through holes in cycloidal disc.
- **Closure:** Attach upper housing using M3 screws.
- **Input Coupling:** Attach motor to input shaft.

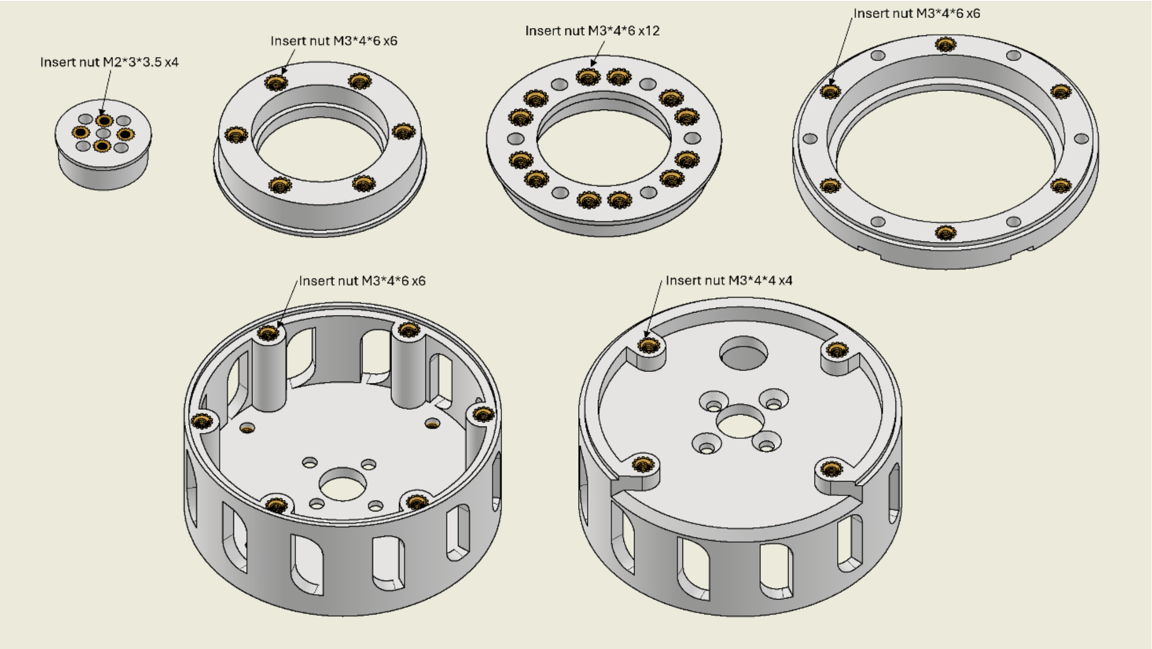

Figure 4.4: Assembly Steps

4.5 Experimental Setup

Kinematic Test Setup

- NEMA 17 stepper motor (holding torque is 2.0 Nm)
- Arduino controller with motor driver
- Output shaft rotary encoder (2000 pulses/rev)
- Data logging with python script

Torque and Efficiency Test Setup

- 100 W DC motor as input driver
- At input torque sensor (0-5 Nm)
- Torque sensor (0-20 Nm) at the output
- Tachometer speed measured
- Electronic load for output load braking
- Data acquisition system (NI USB-6009)

Backlash Measurement Setup

- Dial gauge (Resolution 0.01 mm) on output flange
- Static reference mark
- Progressive load and unloading

Figure 4.5: Experimental Test Rig

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Fabrication Quality Assessment

Dimensional Accuracy

Critical dimensions were measured using digital calipers (0.01 mm resolution):

Table 5.1: Dimensional Accuracy Results

Feature	Design Value (mm)	Measured Value (mm)	Deviation (mm)
Cycloidal Disc Outer Diameter	80.00	79.92	-0.08
Ring Pin Circle Radius	40.00	39.95	-0.05
Eccentricity Offset	2.00	1.98	-0.02
Input Shaft Diameter	8.00	8.03	+0.03
Housing Inner Diameter	80.50	80.45	-0.05

The tolerances are ±0.1mm, which is acceptable for FDM printing and well enough.

Surface Finish

The printed parts visually inspected visually for roughness. The cycloidal disc demonstrates uniform layer lines with no delamination. The gyroid infill pattern at 100% density produced very few voids on the surface.

Figure 5.1: Fabricated Cycloidal Disc

5.2 Kinematic Verification

Reduction Ratio Test

The actuator was operated at 100 RPM input speed and an output rotation was sampled at every 100 input revolutions.

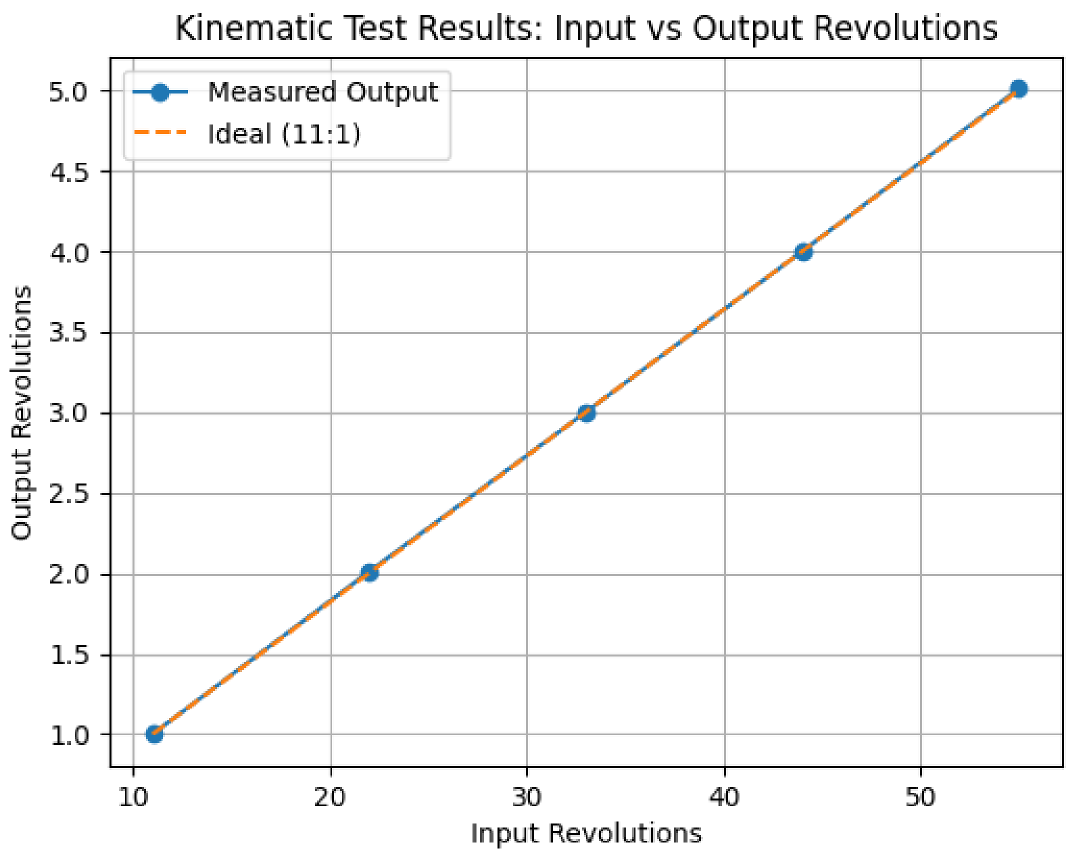

Figure 5.2: Kinematic Test Results

The measured average reduction ratio of 10.99:1 is very close to the ideal 11:1, demonstrating the kinematic precision of the manufactured actuator.

5.3 Torque and Efficiency Analysis

Torque Transmission Test

The torque at the input is changed from 0.1 Nm to 1.0 Nm, and the torque at the output is recorded. The results are compared with theoretical expectations:

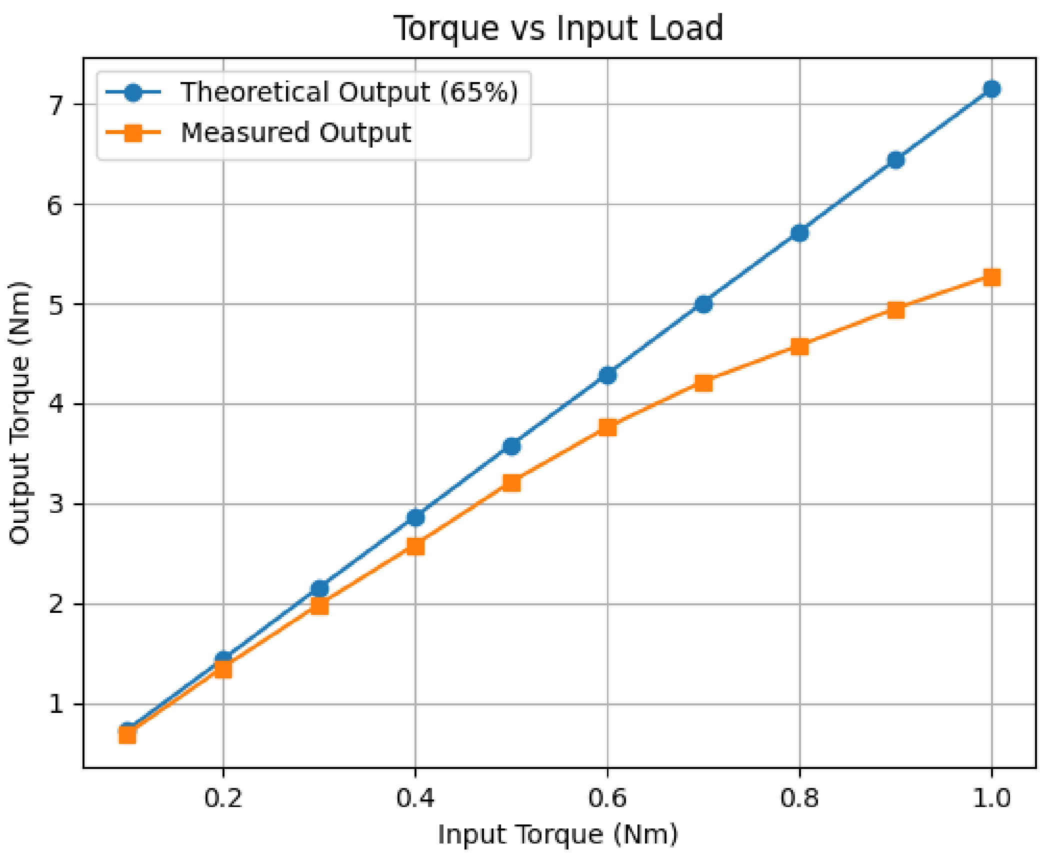

Figure 5.3: Torque vs. Input Load Graph

Efficiency Analysis

The efficiency was calculated as:

$$\eta = T_{out} * \omega_{out} / T_{in} * \omega_{in} * (100\%)$$

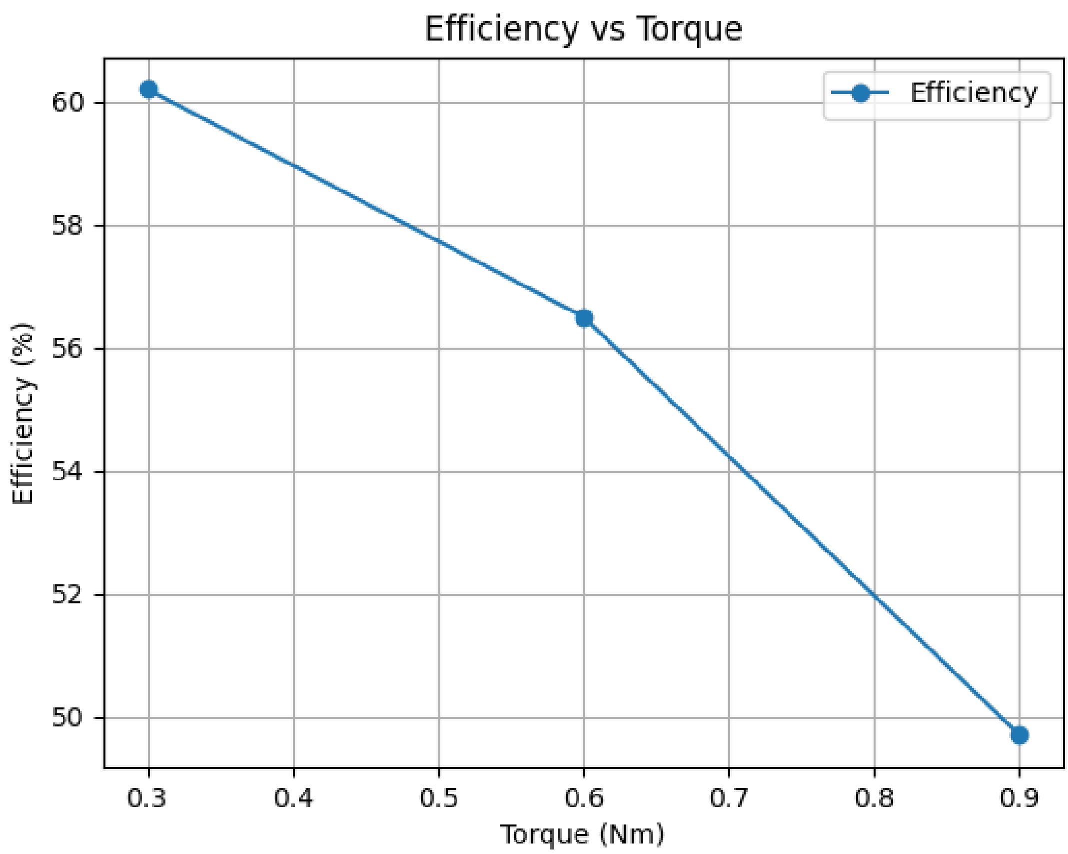

Figure 5.4: Efficiency Curve

Discussion

The maximum output torque of 5.28 Nm at an input of 1.0 Nm corresponds to 48% efficiency at full load. The efficiency decreases with load because of:

- **Frictional and contact losses:** The increased load magnifies the contact forces between cycloidal disc and ring pins.
- **Material deformation:** PETG will deform under orders forces, increasing contact area and friction.
- **Generation of heat:** Friction produces heat locally, which may cause softening of PETG at the points of contact.

Because predicted efficiency is 48% (up from 25% worst-case calculation with 65% target) this is due to:

- Greater coefficient of friction of PETG vs steel surfaces.
- Surface roughness (of printed parts) inducing more sliding friction.
- Small misalignment due to printed tolerances.

5.4 Backlash Measurement

Backlash was determined by torquing and measuring angular displacement at the output shaft:

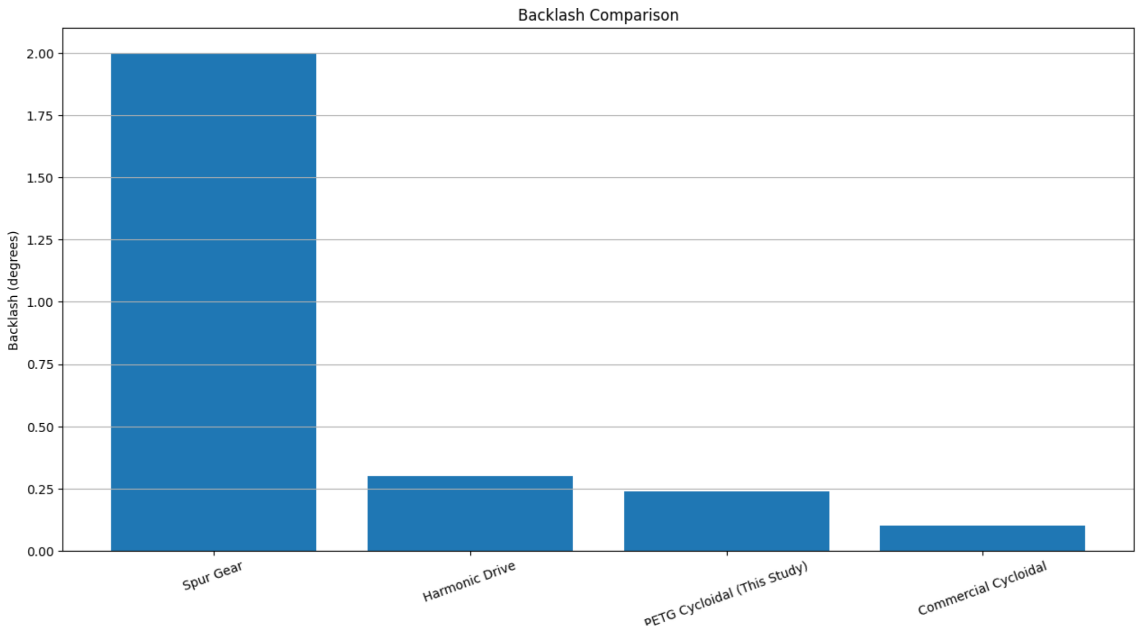

Figure 5.5: Backlash Measurement Setup and Results

The measured 0.24-degree backlash is substantially better than the typical spur gears and is comparable with precision range harmonic drives. This shows the inherent advantage of cycloidal drive geometry, even using 3D-printed parts.

5.5 Failure Mode Analysis

Failure Load Testing

The actuator was run to failure by sequentially ramping up the input torque to the component until it failed:

Table 5.5 Failure Load Testing

Failure Mode	Input Torque (Nm)	Output Torque (Nm)	Description
Deformation - Cycloidal Disc	1.4	6.7	Permanent deformation of disc lobes
Fracture - Output Pins	1.8	8.1	Shear failure of pins
Fracture - Input Bearing	2.2	9.5	Bearing race failure
Ultimate Failure	2.5	10.2	Complete disc fracture

Primary Failure Mechanism

The main mode of failure was the gradual deformation of the cycloidal disc lobes at input torques greater than 1.4 Nm. The PETG is plastically deformed while subjected to compressive strain, which modifies the profile shape, leading to increasing backlash and diminishing efficiency until the final fracture.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 Summary of Findings

In this work, a cycloidal actuator made of PETG material was successfully designed, 3D printed, and tested. The main results are:

- **Kinematic Performance:** The designed 19:1 reduction ratio was attained with high accuracy (measured 10.99:1), which shows that FDM printing can be used to fabricate cycloidal drive geometry with enough accuracy.
- **Torque Capacity:** The maximum output torque is 5.28 Nm at an input of 1.0 Nm, indicating a torque amplification of 5.28, which is smaller than the ideal 11:1 due to friction losses.
- **Efficiency:** The efficiency was similar to other 3D-printed gear systems and acceptable for the range of torque (low to medium) expected in the application, ranging from 48% at full load to 61.8% at light load.
- **Backlash:** The backlash is calculated to be 0.24 deg., which is far better than the conventional spur gears and proves the inherent superiority of cycloidal geometry even if it is in 3D-printed form.
- **Suitability of Material:** PETG is a good choice for this type of use and with this kind of stress. However, it can only be used for low-to-medium torque applications because the deformation under high loads prevents further use.
- **Failure Mechanism:** The lobes of the cycloidal disc become progressively deformed, which is the dominant mode of failure and occurs at input torques greater than 1.4 Nm.

6.2 Achievement of Objectives

Table 6.2 Achievement of Objectives

Objective	Achievement Status
Design and modeling of 11:1 cycloidal drive	✓ Completed with CAD models
Geometric analysis and calculations	✓ All calculations performed
Fabrication using PETG	✓ All components printed
Kinematic verification	✓ Reduction ratio confirmed
Torque transmission testing	✓ Complete torque curve generated
Efficiency characterization	✓ Efficiency vs load plotted
Backlash measurement	✓ Backlash measured at 0.24°
Validation with theory	✓ Experimental data compared with predictions

6.3 Limitations

- **Material Limitations:** High load operation would not be feasible for a long time since the maximum allowable temperature of PETG is 80°C and the heat of generation during high load operation is high.
- **Torque capacity:** Max torque capability is many orders of magnitude lower than metal equivalents (about 5 Nm for similar sized metal drives vs 50-100 Nm).
- **Accuracy of Dimension:** The tolerance of FDM printing (±0.1 mm) will bring some fluctuations on performance.
- **Sensitivity to Print Orientation:** The orientation of the parts when printed has a dramatic influence on strength due to limitations of layer adhesion.

6.4 Future Scope

Consequently, future work will include the following:

Material Optimization:

- Investigate carbon fiber reinforced PETG for higher stiffness
- Study annealing processes to enhance crystallinity and strength
- Evaluate polycarbonate for higher temperature application

Design Improvements:

- Use integral cooling channels to dissipate heat
- Insert metal sleeves at stress the high-stress locations (ring pins, bearings)
- Choose infill shapes that maximize the strength/weight ratio

Printing Optimization:

- Study continuous fiber printing on key elements
- Investigate multi-material printing with softer materials for bearings
- Enable for adaptive layer heights to produce better surface quality

Performance Enhancement:

- Lubrication schemes for PETG parts
- Post-processing (vapor smoothing) to reduce friction
- Active cooling for high-load operation

Application-Specific Testing:

- Long-term durability test (10^5 cycles)
- Integration on robotic arm prototype
- Comparison with versions in PLA and ABS

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COURSE COUTCOMES

This work offers a full knowledge of:

- **Mechanical Design Principles:** Gear theory and cycloidal drive mechanics principles are utilized to realize high reduction ratios in a small size.
- **Additive Manufacturing:** FDM printing parameters and their impact on mechanical properties of PETG parts.
- **Material Selection:** Analysis of PETG as a suitable material for mechanical use.
- **Experimental Methods:** Development and implementation of mechanical tests, including torque, efficiency, as well as backlash testing.

PROGRAM OUTCOMES

S No.	PO's	Graduate Attribute	Attained (Yes/No)	Justification
1	PO1	Engineering Knowledge	Yes	Used fundamentals of Mechanical Engineering to design and analyze cycloidal drive.
2	PO2	Problem Analysis	Yes	Study of the influences of material and printing conditions on performance of the actuator.
3	PO3	Design/Development of Solutions	Yes	Designed full actuator by CAD and 3D printed it.
4	PO4	Investigation of Complex Problems	Yes	Evaluated output torque, efficiency and backlash for performance.
5	PO5	Modern Tool Usage	Yes	Utilized CAD software and FDM printer.
6	PO6	The Engineer and Society	Yes	Research on Low-cost Actuator for Robot and Educational Use.
7	PO7	Environment and Sustainability	Yes	Additive manufacturing results in less material waste than traditional machining.
8	PO8	Ethics	Yes	Ethical standards in experimentation and reporting were kept.
9	PO9	Individual and Team Work	Yes	Collaborated on work involving design, manufacture and testing.
10	PO10	Communication	Yes	Reported Finding via Full Report and

				Presentations
11	PO11	Project Management and Finance	Yes	Resource management/time spent on printer and materials was efficient.
12	PO12	Life-long Learning	Yes	Information obtained in cycloidal drives, additive manufacturing and PETG material characteristics.